

1M1B – AI for Sustainability Virtual Internship
(In Collaboration with IBM SkillsBuild & AICTE)

Final Project Submission

Gomti Saathi: A Simple AI Assistant That Helps Lucknow Residents Understand River Water Quality

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Primary SDG: SDG 6 – Clean Water and Sanitation
Secondary SDG: SDG 11 – Sustainable Cities and Communities

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1. Project Description

1.1 How I arrived at this problem

I live in Lucknow, and the Gomti river runs right through the middle of the city. You cannot grow up here without noticing it. Near Gaughat the water looks dull in summer, and once you move downstream towards the Nishatganj and Kudiyaaghat drains it gets visibly worse – foam on the surface, a smell, garbage stuck at the edges. People still bathe in some stretches, children play near the ghats, and washermen use the water daily.

While looking for a project topic for this internship, I checked whether any data on this exists. It does. The Uttar Pradesh Pollution Control Board (UPPCB) tests the river every month at fixed stations and publishes advisories, and the CPCB classifies water quality into categories A to E based on parameters like dissolved oxygen (DO), BOD, pH and coliform counts. A recent advisory, for example, recorded acceptable DO and BOD at the upstream Manjhi ghat station, while several city stretches such as Kudiyaaghat, Nishatganj bridge and Pipraghat were placed in category E, the lowest class, fit only for controlled waste disposal.

This is where the real gap is. The data exists, but it sits in government PDFs and newspaper reports written in technical language. The people who live next to the river have never seen it, and even if they did, words like “faecal coliform” or “BOD 6.2 mg/l” mean nothing without context. So the problem is not a shortage of information. It is that the information never reaches ordinary people in a form they can act on.

1.2 Problem statement

How might we use AI to convert publicly available river water quality data into simple, location-specific answers, so that residents of Lucknow can make safer decisions about the Gomti river and participate in its protection?

1.3 SDG alignment

The project mainly supports SDG 6 (Clean Water and Sanitation), because it deals directly with water quality information and public health around a water body. It also touches SDG 11 (Sustainable Cities and Communities), since the Gomti is part of Lucknow's living environment, and informed citizens are the first step towards a cleaner urban river. I have kept SDG 6 as the single main focus, as the guidelines suggest.

1.4 AI solution overview

My proposed solution is a chatbot called Gomti Saathi (“saathi” means companion). It is built as a Retrieval-Augmented Generation (RAG) system using the IBM Granite models that we were introduced to during the internship. The idea is deliberately simple:

- A small knowledge base is created from public documents: UPPCB monthly water quality advisories for the Lucknow stations, CPCB classification criteria and permissible limits, and a short list of common citizen questions (is it safe to bathe, wash, fish, let children play near the water).
- When a user asks a question, the system first retrieves the relevant pieces of these documents, then passes them to the model along with a strict system prompt.
- The model answers in plain language, names the station and the month the data belongs to, and gives a clear verdict (for example: “not safe for bathing”) with the reason in one line.

No advanced coding is needed for this. The core work is in prompt design, document chunking and retrieval logic, which is exactly what the internship trained us on. I built the prototype workflow on paper and tested the prompt logic with sample questions, as shown in Section 3.

1.5 Target users

- Residents living near the Gomti, especially around the ghat areas from Gaughat to Pipraghat.
- People who bathe, wash clothes or fish in the river, and parents of children who play near the water.
- School and college students and small local NGOs who run river clean-up and awareness activities.
- At a later stage, municipal or UPPCB staff could also use it to see what citizens are asking about which locations.

1.6 Responsible AI considerations

Fairness: The assistant must serve people who do not read English well, so answers are kept in simple language and the design allows Hindi or Hinglish questions. Verdicts are based only on published official data, not on assumptions about a particular area or community, so no locality gets unfairly labelled as “dirty” without a source.

Transparency: Every answer states where the information comes from (which advisory, which month, which monitoring station). The system prompt instructs the model to never invent numbers. If the retrieved data is old or missing, the bot must say so clearly instead of guessing.

Ethics: A wrong “safe” answer here could harm someone's health, so the bot is instructed to be cautious: when in doubt, it advises against contact with the water and suggests checking the latest UPPCB report. It is an information tool, not a replacement for official testing.

Privacy: The chatbot does not ask for names, phone numbers or locations beyond the river stretch being discussed. No personal data is stored or used, and all source documents are already in the public domain.

1.7 Expected impact

- **Health and safety:** people avoid bathing or washing in stretches that are officially in category D or E, and understand why, instead of relying on guesswork.
- **Awareness:** the monthly data that usually stays inside government reports reaches ordinary residents in a conversation they can understand.
- **Accountability:** when citizens know which drains and stations are failing, they can raise specific complaints and cite actual numbers.
- **Reusability:** the same RAG pattern works for any river and any city. You only replace the documents in the knowledge base, so a student in Kanpur or Varanasi can copy the whole workflow for their own river.

2. Design Thinking Process

The internship asked us to follow the five design thinking stages, so I have kept my notes in the same order here. It genuinely helped, because it slowed me down before I jumped into building anything.

2.1 Empathize

Instead of starting on the internet, I spent a few evenings near the river to understand who actually faces this problem. I spoke informally with six or seven people around Gaughat and Kudiyaaghat: two washermen who use the river water every day, a tea seller whose stall stands near the ghat steps, one fisherman, and a few residents who come for morning walks. What I heard again and again was not a complaint about pollution itself. Everyone already knows the river is dirty. The confusion was about how dirty, and where. One washerman told me he avoids the stretch near the drain “on some days” but could not explain which days or why. A resident believed the water becomes “fine after the monsoon rains”, which had no basis at all. Not one of them had ever heard of the UPPCB monthly advisory.

This visit changed my understanding of the problem. The pain point is not general awareness of pollution. It is the absence of usable, location-specific information at the moment a person decides whether to enter the water or not.

Why does the problem persist? Three reasons came out of those conversations. First, the reports are in technical language and PDF format. Second, there is no local channel that converts them into everyday advice. Third, people trust their own eyes and old habits more than documents, so unless the data is explained simply, it will never beat habit.

2.2 Define

From those observations I wrote the problem statement given in Section 1.2. The target users are listed in Section 1.5, with residents of the ghat areas as the primary group. The current gaps I could define clearly are:

- Data is published but never explained. There is a gap between a government report and a citizen's understanding of it.
- Advice is not location-specific. The Gomti is one river, but its quality changes stretch by stretch, and a single city-wide statement hides that.
- There is no free question-and-answer channel in simple language. A resident cannot call anyone and ask about their own ghat.
- Existing awareness runs on appearance and rumour, which is why beliefs like “clean after rains” survive.

2.3 Ideate

I brainstormed four possible solutions and wrote down honestly why I accepted or rejected each one:

- Low-cost water sensors with an IoT dashboard. Rejected. Hardware cost, calibration and maintenance are far beyond a student project, and official lab data already exists every month.
- A static website that summarises monthly reports. Rejected. Pages go stale, nobody visits them regularly, and it cannot answer a specific question about a specific ghat.
- Image classification of river photos to judge safety. Rejected for now. Murky water can be safe-ish and clear water can be unsafe, so a photo-based verdict could mislead people. Kept only as a possible

future add-on for reporting foam or garbage.

- A chatbot with RAG over official documents. Selected. It reuses data that is already public, answers location-specific questions in simple language, works on an ordinary phone, and uses exactly the skills taught in this internship: prompt engineering, retrieval and IBM Granite models.

Writing this comparison was useful in itself. It taught me to pick the simplest tool that answers the user's question honestly, instead of the most impressive one.

2.4 Prototype

The prototype was built on paper first, before any software. I wrote down twelve questions a resident might realistically ask, then for each question I manually picked the paragraph from the advisories and CPCB criteria that should be used to answer it. That manual exercise became my retrieval logic. The flowchart, the system prompt and the sample question-answer pairs in Section 3 are the final prototype output of this stage.

2.5 Test and refine

I showed the paper prototype and sample answers to three people: a senior who has done a similar internship before, a friend studying environmental science, and my uncle who lives near Pipraghat and represents the real user. Their feedback directly changed the design:

- My uncle said the first answer format was too long and full of terms. I changed the format to verdict first, then at most two lines of plain reason.
- My environmental science friend pointed out that monthly data can mislead someone mid-month. This became Rule 3 of the system prompt: always state the month, and warn if the data is older than two months.
- My senior asked what stops the model from inventing a number. I strengthened Rule 1 and made the station name and month mandatory in every answer, so a fabricated figure would be easy to catch.
- One tester typed his question in Hindi without telling me. That made me add Rule 6, answering in the user's own language.

These small corrections convinced me that responsible AI is not a section you write at the end of a report. It enters the design through testing with real people.

3. Prototype and Demo

3.1 Overall workflow

The diagram below shows the complete flow of a single question, from the user to the final answer. It was drawn by hand first and then recreated here.

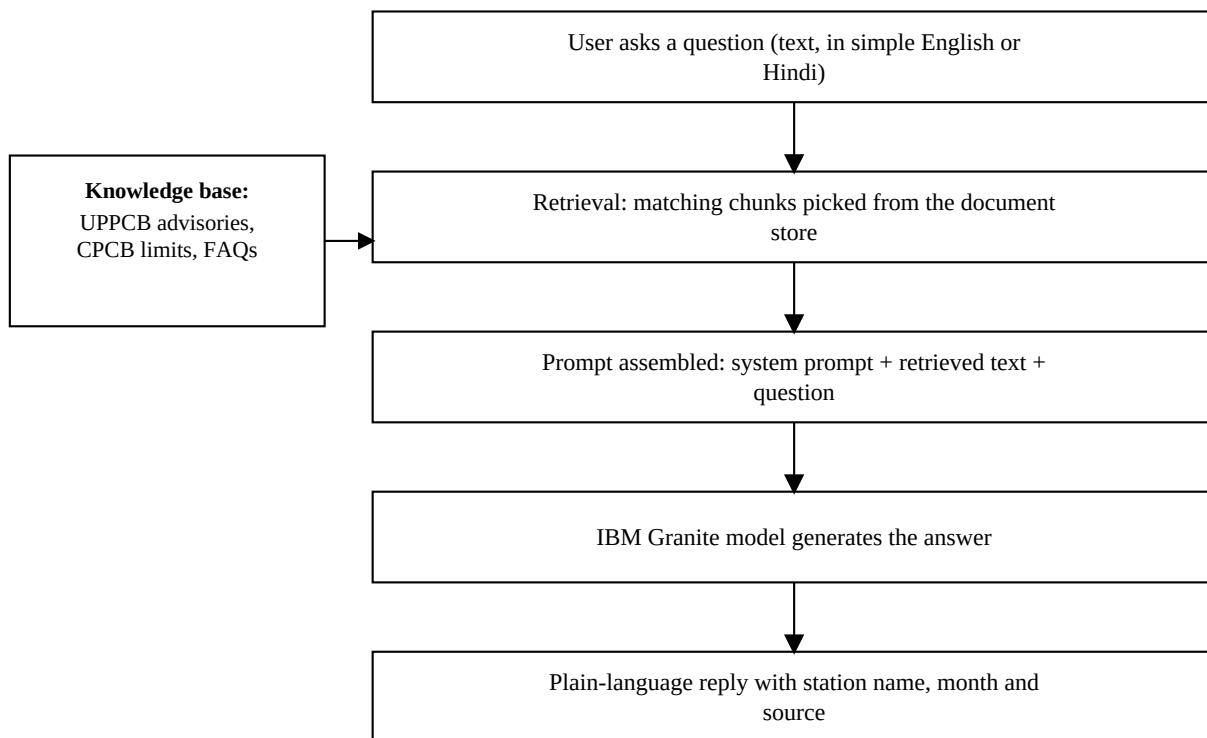


Figure 1: Working of Gouti Saathi for one user question.

3.2 Contents of the knowledge base

The knowledge base is intentionally small. Four document sets are enough, and all of them are already public:

Document	What it contains	What it is used for
UPPCB monthly water quality advisories (Lucknow stations)	Station-wise DO, BOD, pH and coliform values with the month of sampling	The numbers and the safety verdict in each answer
CPCB water quality classification criteria	Definition of categories A to E and their designated uses	Explaining what a category means in plain words
CPCB permissible-limit summaries	Acceptable ranges for bathing, washing and similar contact uses	Converting raw readings into safe or not safe
My own FAQ notes from field visits	Twelve typical citizen questions with simple explanations	Matching a casual question to the right document chunk

Table 1: Documents kept in the knowledge base and their role.

3.3 System prompt used in the prototype

This is the instruction set I wrote for the model. It is the most important part of the prototype, because it controls honesty and safety of the answers:

You are Gomti Saathi, a helpful assistant about the water quality of the Gomti river in Lucknow.

Rules:

1. Answer only using the provided documents. Never invent readings, dates or station names.
2. Always mention the monitoring station and the month the data belongs to.
3. If the data is older than two months or the station is not covered, say so clearly and suggest checking the latest UPPCB advisory.
4. Give a clear verdict first (safe / not safe / avoid contact), then explain in one or two simple lines. Avoid technical words unless you explain them.
5. If unsure, always advise caution. Do not encourage bathing or washing in category D or E stretches.
6. Answer in the same language the user writes in (simple English or Hindi).

3.4 Sample inputs and outputs

I tested the prompt logic with typical questions a resident might ask. The answers below are what the workflow produces when the relevant advisory text is supplied as context. Numbers quoted are from published UPPCB reports for illustration.

Q1: Can we bathe in the Gomti near Gaughat these days?

A: Based on the latest UPPCB advisory available in my records, the upstream stations near the city show better water (for example, DO around 8.4 mg/l and BOD around 2.9 mg/l at Manjhi ghat), which is acceptable for outdoor bathing. However, inside the city several stretches like Kudiyaighat and Nishatganj bridge are in category E, so bathing there should be avoided. Please check which exact ghat you mean, and note that this data is monthly, so it can change.

Q2: What does category E mean?

A: The CPCB ranks river water from A to E. Category E is the lowest class. It means the water is fit only for controlled waste disposal, not for bathing, washing or any contact use. It usually indicates low dissolved oxygen and high pollution load, often from untreated sewage.

Q3: There is foam near Nishatganj bridge every morning. Is this normal?

A: Foam like this usually comes from detergents and sewage entering through the drains. Detergents add phosphates, which feed algae growth; when algae die and decompose, oxygen in the water drops further. This stretch is already classified in category E in recent advisories, so contact with the water should be avoided, and the foam can be reported to the pollution board with the location and date.

Q4: My children play near Gomti barrage. Should I stop them?

A: Recent advisories place the Gomti barrage stretch in category E, which is not fit for any contact with water. So yes, children should be kept out of the water itself. Playing on the bank is less risky, but they should wash their hands properly afterwards, and balls or toys that fall into the water should not be retrieved by hand.

Q5: The river looks much cleaner after rains. Does that mean it is safe now?

A: Not necessarily. Rain dilutes the pollution, so the river can look and smell better for a while, but bacteria and coliform levels can still be unsafe, and drains keep discharging through the season. Appearance is not a test. Only the lab readings in the monthly advisory can say whether a stretch is safe, so please check the latest report for your station before using the water.

3.5 Tools and skills used

- Prompt engineering for the system prompt and answer format.
- RAG pipeline design (document collection, chunking, retrieval).
- IBM Granite models, as introduced in the internship modules.
- IBM SkillsBuild course material on responsible AI.

4. Impact Statement

If Gomti Saathi were actually implemented for Lucknow, the first change would be very small but practical: families living near the river would know, in one question, whether this week's water at their ghat is safe for contact. Right now they judge by appearance and habit. Converting that judgement into a data-backed answer reduces health risks like skin infections, which are common among people who bathe in polluted stretches.

The second change would be in awareness. Most residents do not know that monthly official testing even happens. A simple bot that cites the station and month makes that data visible. Visible data tends to create public pressure, and public pressure is what pushes cleanup work forward. Local NGOs and student groups could use the same tool in their campaigns instead of preparing their own technical summaries.

Who benefits and how:

- Residents near the ghats: safer daily decisions about bathing, washing and fishing.
- Children and daily wage workers who spend time at the river: fewer exposure risks.
- NGOs and awareness groups: ready-made, sourced information for campaigns.
- Pollution board and city authorities: an additional channel showing which locations citizens are worried about.

The cost of running such a tool is low because it only uses documents that are already public, and the same design can be repeated for any Indian river by changing the knowledge base. That is what makes this project sustainable in the literal sense: small, reusable and honest about its limits.

5. Limitations and Future Scope

No honest project report is complete without its weak points, and this one has several.

- The system depends on monthly official data. A sudden industrial spill between two advisories will not be reflected until the next report, and the bot must say so rather than pretend otherwise.
- The answers are only as good as the documents. If UPPCB changes the report format, the chunks and retrieval logic will need fixing.
- It is a text-based tool, so it excludes people without smartphones or without literacy. This is a real limitation I could not solve within this project.
- What I built is a conceptual prototype. It has not been deployed, so I have not seen how real users behave with it over time.

If I continue this work after the internship, the next steps I would try are:

- A voice interface in Hindi, so that non-literate users can ask and listen instead of reading.
- Hosting it on WhatsApp or SMS, because that is where people already are, instead of expecting them to install a new app.
- Repeating the same pipeline for the Sai river and other stretches in Uttar Pradesh by simply replacing the knowledge base documents.
- Allowing citizens to send dated photos of foam or garbage as pollution reports to the board. These would be reports only, never safety verdicts, to avoid the misleading photo-classification risk mentioned in Section 2.3.

6. What I Learned During This Project

Before this internship I used to think an AI project has to involve training a big model or writing a lot of code. Working on Gomti Saathi changed that view. The hardest part was not the technology; it was defining the problem correctly and making sure the system would not mislead people when data was missing. Learning to write strict prompts, to ground answers in documents, and to think about fairness and privacy from the beginning was the most useful part of the whole exercise.

The field visits taught me something else too. The washerman at Gaughat does not need a research paper. He needs one line he can trust. Keeping that sentence in my head made every design decision in this project easier.

I would like to thank the 1M1B team, IBM SkillsBuild and AICTE for this opportunity, and my mentors for their feedback during the internship sessions.

References

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- [3] Times of India, Lucknow edition, report on UPPCB advisory: “Gomti water quality poor in city due to sewage discharge”.
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